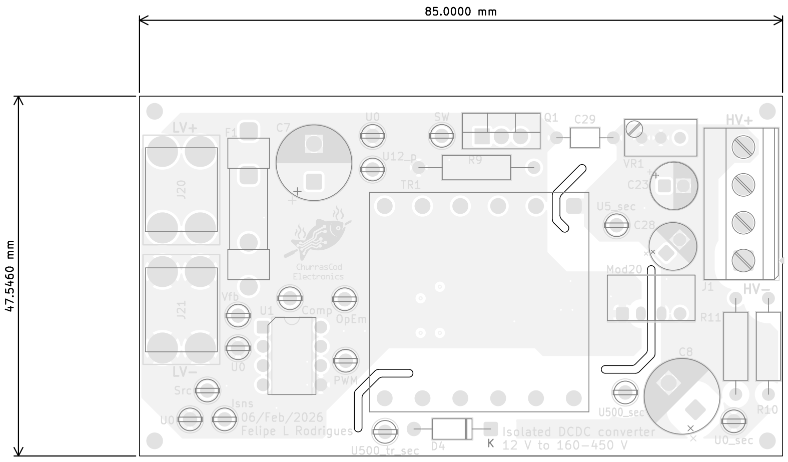
Step-Up Fabrication Document

Top Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.

4) SURFACE FINISH: IMMERSION GOLD

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR YELLOW.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

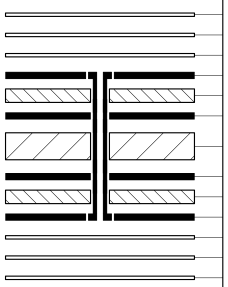
9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY FR.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	85.000 × 47.546 mm
BOARD THICKNESS	1.602 mm
TRACE WIDTH	0.400 mm
TRACE TO TRACE	-0.000 mm
MIN. HOLE (PTH)	0.250 mm
MIN. HOLE (NPTH)	2.100 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.200 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm

Layer Stack Legend



Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Ref)**	**0.07mm (2.00oz)**		**Plane**	**GBR**
Prepreg		0.001mm	FR4_7628	Dielectric	
Copper	**Virtual (Top side jumpers)**	**0.035mm (1.00oz)**		**Copper Layer**	**GBR**
Core		1.35mm	FR4	Dielectric	
Copper	**Virtual (Bot side jumpers)**	**0.035mm (1.00oz)**		**Copper Layer**	**GBR**
Prepreg		0.001mm	FR4	Dielectric	
Copper	**L2 (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.602mm
Note: external layer thicknesses are specified after plating

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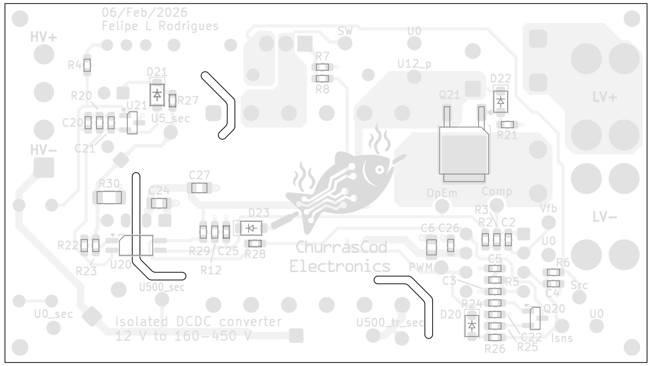
Comments:	Company: null		Variant: PRELIMINARY	Git Hash: 5fb04d2
	Board Name: **Step-Up**		Project Name: **Isolated step-up 12:450 V**	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer: FR	Size: **A4**	Sheet: **1** of **7**

Step-Up Fabrication Document

Bottom Fabrication (Scale 1:1)



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Comments:	Company: null		Variant: PRELIMINARY	Git Hash: 5fb04d2
	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer: FR	Size: A4	Sheet: 2 of 7

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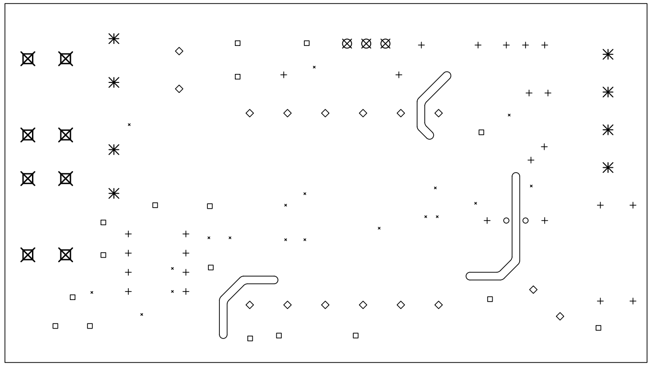
Step-Up Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	19	0.25mm (9.84mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
○	2	0.70mm (27.56mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
+	25	0.80mm (31.50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
□	17	0.90mm (35.43mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
◇	16	1.00mm (39.37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊠	3	1.20mm (47.24mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
✳	8	1.30mm (51.18mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊠	8	1.80mm (70.87mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
	Total 98					



ChurrasCod Electronics	Comments:	Company: null	Variant: PRELIMINARY	Git Hash: 5fb04d2
		Board Name: **Step-Up**	Project Name: **Isolated step-up 12:450 V**	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 1.0.0+ (Unreleased)
	Sheet Path:		Reviewer: FR	Size: A4	Sheet: 3 of 7

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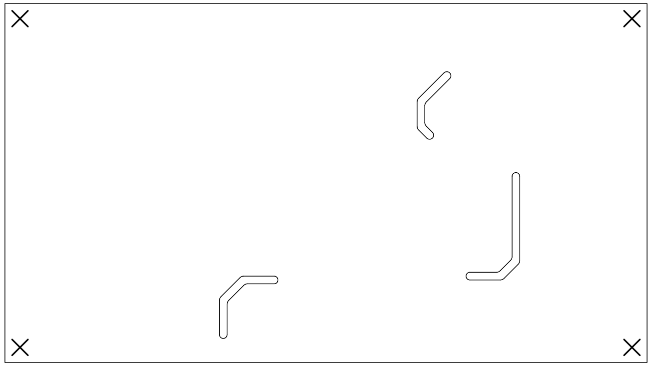
Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	2.10mm (82.68mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
	Total 4					



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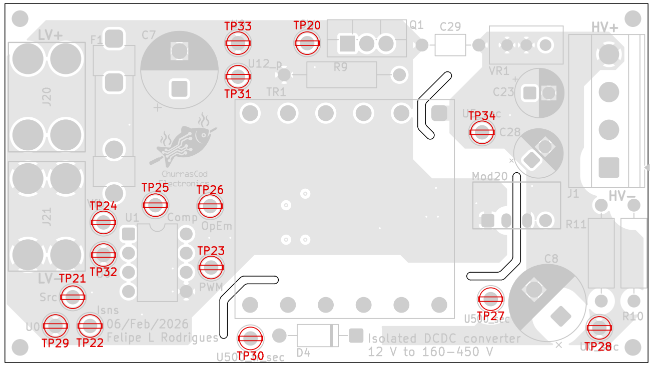
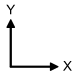
			Revision: 1.0.0+ (Unreleased)
Sheet Path:	Reviewer: FR	Size: A4	Sheet: 4 of 7

Step-Up Fabrication Document

Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP20	SW	41.95	44.73
TP21	Src	10.97	11.07
TP22	Isns	13.25	7.26
TP23	PWM	29.25	15.01
TP24	Vfb	15.03	20.98
TP25	Comp	21.89	23.27
TP26	OpEm	29.13	23.14
TP27	U500_sec	66.21	10.82
TP28	U0_sec	80.56	7.01
TP29	U0	8.68	7.26
TP30	U500_tr_sec	34.46	5.61
TP31	U12_p	32.81	40.28
TP32	U0	15.03	16.66
TP33	U0	32.81	44.73
TP34	U5_sec	65.07	32.92





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	Board Name: **Step-Up**			Project Name: **Isolated step-up 12:450 V**	
Sheet Title: Top Test Points (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR		Date: 2026-02-07	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer: FR		Size: A4	Sheet: 5 of 7

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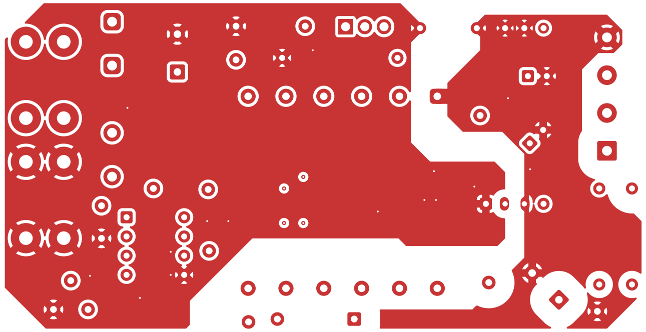
Electronics

Step-Up Fabrication Document

F.Cu (Scale 1:1)



Comments:	Company: null		Variant: PRELIMINARY	Git Hash: 5fb04d2
	Board Name: **Step-Up**		Project Name: **Isolated step-up 12:450 V**	
Sheet Title: F.Cu (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer: FR	Size: **A4**	Sheet: **6** of **7**

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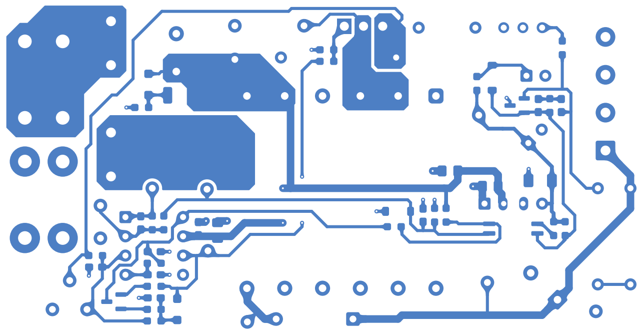
Step-Up Fabrication Document

B.Cu (Scale 1:1)



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Comments:	Company: null		Variant: PRELIMINARY	Git Hash: 5fb04d2
	Board Name: **Step-Up**		Project Name: **Isolated step-up 12:450 V**	
Sheet Title: B.Cu (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer: FR	Size: **A4**	Sheet: **7** of **7**